APPSC INTEGRATED ENGINEERING INDEX & MICRO-TOPIC OVERLAP ANALYZER

14–17 & 23. SOM, FLUID MECHANICS, MATERIALS & PROCESS THERMODYNAMICS

Underlined Highlights: **AEE Common P-II (Civil/PR) Extras** | **AEE PCB P-II Extras** | **AEE Civil P-III (U6) Extras**

AEE COMMON P-II: CIVIL & PR/TW (Units 1 & 2 | 150M) **AEE PCB P-II: (Units 4, 5 & 6)**

AEE CIVIL P-III: UNIT 6 (Hydraulics & Water Resources)

Subj	Consolidated Engineering Subject	Exams & Notification Clause Mapping	Overlap Type	Page
14A	**Engineering Mechanics, Stresses, SFD/BMD & Bending**	AEE Common P-II Unit 1 • AEE PCB P-II Unit 4 (Stress-Strain)	100% Civil + PR Common	**01–02**
14B	**Principal Stresses, Columns, Torsion, Springs & Shells**	AEE Common P-II Unit 1 • AEE Civil P-III Unit 2 (Link)	100% Civil + PR Common	**02**
15A	**Fluid Statics, Kinematics, Dynamics & Pipe Flow**	AEE Common P-II U2 • AEE PCB P-II U5 • AEE Civil P-III U6	Triple Overlap (Civil+PR+PCB)	**03**
15B	**Hydraulic Turbines, Pumps, Valves & Hydro Plants**	AEE Common P-II U2 • AEE Civil P-III U6 (Hydrology Link)	Civil + PR Only (Absent in PCB!)	**04**
23	**Open Channel Flow, Similitude, Hydrology & Irrigation**	AEE Civil P-III Unit 6 • AEE Common P-II U2 • AEE PCB U5	Similitude/Hydro Overlap + Civil Design	**05**
16–17	**Engineering Materials & Process Thermodynamics (PCB)**	AEE PCB Paper-II Unit 4 & Unit 6	100% Exclusive to AEE PCB	**06**

SUBJECT 14 • STRENGTH OF MATERIALS & ENGINEERING MECHANICS (PART A & B)

AEE COMMON P-II UNIT 1 (AEE CIVIL + AEE PR/TW) | PCB P-II UNIT 4 LINK

1. Strength of Material (AEE Notification 27/2026 — Common Paper-II for Civil & PR/Tribal Welfare)

AEE CIVIL + AEE PR P-II (75M)

Mechanics & Simple Stresses: Forces, moments, Equilibrium; Applying the Equation of Equilibrium, Planar Trusses; Friction. Simple Stresses & Strains: Elasticity and plasticity, Types of stresses & strains, Generalized Hooke's law – **stress-strain diagram for mild steel** – Working stress – Factor of safety – Lateral strain, Poisson's ratio & volumetric strain – Elastic moduli & the relationship between them – Bars of varying section – composite bars – Temperature stresses. Strain energy – Resilience – Gradual, sudden, impact and shock loadings.

SFD, BMD, Flexural & Shear Stresses: Definition of beam – Types of beams – Concept of shear force and bending moment – S.F. and B.M. for cantilever, simply-supported and overhanging beams subjected to point loads, U.D.L., Uniformly varying loads and combination of these loads – Point of contraflexure – Relation between S.F., B.M. and rate of loading at a section of a beam. Flexural Stresses: Theory of simple bending – Assumptions – bending equation: Neutral axis – bending stresses – section modulus of different sections – Design of simple beam sections. Shear Stresses: Derivation of formula – Shear stress distribution across various beam sections.

1. Strength of Material (Continued — Principal Stresses, Columns, Torsion, Springs & Shells) AEE CIVIL + AEE PR P-II

Principal Stresses and Strains: Stresses on an inclined section of a bar under axial loading – Compound stresses – Normal and tangential stresses on an inclined plane for biaxial stresses – Two perpendicular normal stresses accompanied by a state of simple shear – Mohr's circle of stresses – Principal stresses and strains – Analytical and graphical solutions. Different theories of Failure: Various theories of failure. Columns and struts: Euler's column theory – types of end conditions; critical load on the column derivations – Rankine's formula for columns. Lifting machines: definitions, Law of machine, study of important lifting machines; virtual work principle. Torsion of Circular Shafts: Theory of pure torsion – Torsion Equations: Assumptions made in the theory of pure torsion – Torsional moment of resistance – Polar section modulus – Power transmitted by shafts – Combined bending and torsion and end thrust. Springs – Helical and leaf springs. Thin & Thick Cylinders and Spherical shells: Thin seamless shells – Formula for longitudinal and circumferential stresses and max shear stresses – hoop, longitudinal and volumetric strains – changes in diameter, and volume of thin shells .

4. ENGINEERING MATERIALS (Stress-Strain Link in AEE PCB Paper-II) AEE PCB PAPER-II (U4)

Structure and properties of engineering materials, phase diagrams, heat treatment, stress-strain diagrams for engineering materials.

SOM Sub-Module	100% Common for AEE Civil + AEE PR/TW (Paper-II Unit 1)	Overlap with AEE PCB & AEE Civil Paper-III
1. Stress-Strain & Elastic Constants	• Hooke's Law, Stress-Strain curve for mild steel & brittle/ductile materials • 4 Elastic constants (E, G, K, μ) & inter-relations • **Composite bars, Thermal stresses, Impact loading & Strain energy**	• **AEE PCB U4 Overlap:** Stress-strain diagrams for engineering materials (Yield point, UTS, Proof stress, Ductility, Toughness, Resilience). • **AEE Civil P-III U2 Overlap:** Strain energy & Virtual Work principle recur in Structural Analysis!
2. Statics, Trusses, Lifting Machines & Friction	• **Equations of Equilibrium (ΣFx=0, ΣFy=0, ΣM=0), Planar Trusses (Method of Joints & Sections), Friction (Wedge/Ladder)** • **Lifting Machines: Mechanical Advantage (MA), Velocity Ratio (VR), Efficiency, Law of Machine (P = mW + C), Reversible vs Self-locking machines, Differential pulley/Screw jack**	• **Absent in AEE PCB!** • Note: **Lifting Machines** is a classic Engineering Mechanics topic inside Paper-II Unit 1 that Civil aspirants often skip—gives 2–3 easy formula questions in both AEE Civil & AEE PR!
3. SFD/BMD, Bending, Shear, Torsion, Columns, Springs & Shells	• **SFD/BMD, Point of contraflexure, Simple Bending (M/I = σ/y = E/R), Shear stress distribution (Rectangular, Circular, I, T, Triangular)** • **Mohr's Circle, 5 Theories of Failure (Rankine, Guest/Tresca, St. Venant, Haigh, Von Mises)** • **Euler's & Rankine's Columns, Shaft Torsion (T/J = τ/R = Gθ/L), Helical/Leaf Springs, Thin & Thick (Lame's) Cylinders/Spherical Shells**	• **Absent in AEE PCB!** • **100% Identical Paper-II for AEE Civil & AEE PR:** Because both posts share the exact same Paper-II exam paper, every single SOM topic carries equal weightage (approx. 75 out of 150 questions in Paper-II).

⚡ **YUDDAM 2027 AUDIT INSIGHT (SUBJECT 14 — SOM):** AEE Civil (R&B/WRD/PH&ME/RWS) and AEE PR/Tribal Welfare have the **exact same Paper-II (150 Marks = 75M SOM + 75M Fluid Mechanics)**. However, AEE PCB does **NOT** have SOM —only the *stress-strain diagram* portion appears in PCB Unit 4. Do not waste time reading SFD/BMD, Torsion, or Columns when revising exclusively for AEE PCB!

SUBJECT 15A • FLUID MECHANICS CORE (TRIPLE OVERLAP: CIVIL + PR + PCB)

AEE COMMON P-II UNIT 2 (PART 1) | AEE PCB P-II UNIT 5

2. Fluid Mechanics and Machinery (Part 1: Fluid Statics, Kinematics, Dynamics, Pipes & Boundary Layer) — AEE CIVIL + AEE PR P-II

Fluid statics: Dimensions and units: physical properties of fluids – specific gravity, viscosity, and surface tension – Vapour pressure and their influence on fluid motion – atmospheric, Pascal's law, gauge and vacuum pressures – measurement of pressure: Piezometer, U-tube and differential manometers. Hydrostatics, Fluid forces on planes and curved surfaces, submerged and floating bodies, Buoyancy and stability.

Fluid kinematics: Description of flow pattern and types of fluid flows – Velocity and acceleration: convective, temporal, tangential and normal accelerations, control volume – basic principles of fluid flow, continuity equation for 3-D, 2-D, 1-D flow. Rotational and irrotational motion, Velocity potential, stream function, flow net.

Fluid dynamics: Surface and body forces – Euler's and Bernoulli's equations for flow along a streamline and its applications, momentum equation and its applications. Flow measurement devices – Gross measurement: Venturimeter, Orificemeter, Turbine flow meters, Rotameters; Pressure measurement: Pitot tubes, Hot wire/film anemometer, their measurement principles and sources of errors; calibration.

Closed conduit flow & Boundary Layer: Reynolds experiment – Major and Minor losses in pipes – pipes in series and pipes in parallel – total energy line – hydraulic gradient line, water hammer. Boundary Layer Concepts: Definition, thicknesses, characteristics along thin plate, laminar and turbulent boundary layers – boundary layer in transition, separation of boundary layer, submerged objects – Drag and lift.

5. FLUID MECHANICS AND HYDRAULICS (AEE Notification 08/2026 — AEE PCB Paper-II Unit 5) — AEE PCB PAPER-II (U5)

Newtonian and non-Newtonian fluids – Fluid Properties; Measurement of Pressure – Manometers; Fluid Kinematics – Classification of Fluids, Stream function and Velocity potential, Fluid dynamics – Continuity equation, Bernoulli's equations and Impulse momentum equation; Laminar and Turbulent flow through pipes – significance of Reynolds number, Hagen-Poiseuille's equation, Darcy-Weisbach equation, Friction factor, head losses in pipes, bends and fittings; Dimensional analysis and similarity laws; elementary boundary layer theory.

100% Common Fluid Core (AEE Civil + AEE PR + AEE PCB)	Extra Sub-Topics ONLY in AEE Common P-II (Civil & PR)	Explicitly Named in AEE PCB U5 (& Civil P-III U6)
• Fluid properties (Viscosity, Surface tension, Capillarity, Compressibility) • Manometers (Piezometer, U-Tube, Differential) • Kinematics: Types of flow, 1D/2D/3D Continuity, Stream function (ψ) & Velocity potential (φ) • Bernoulli's Equation & Impulse-Momentum Equation (Pipe bends/nozzles) • Laminar (Hagen-Poiseuille) & Turbulent Pipe Flow (Darcy-Weisbach, Major/Minor losses) • Elementary Boundary Layer theory	• Hydrostatic Forces on Plane & Curved Surfaces (Total pressure & Center of pressure) • Buoyancy, Metacentric Height & Stability of Submerged/Floating Bodies • Flow Meters & Anemometers: Venturimeter, Orificemeter, Rotameter, Turbine meter, Pitot tube, Hot-wire & Hot-film anemometer • Compound Pipes (Series/Parallel), TEL/HGL, Water Hammer (Joukowsky eq) & Drag/Lift on submerged bodies	• Rheology: Newtonian vs Non-Newtonian Fluids (Pseudoplastic, Dilatant, Bingham Plastic, Thixotropic, Rheopectic) • Dimensional Analysis (Rayleigh's & Buckingham's Pi Theorem) & Similitude Laws (Reynolds, Froude, Euler, Weber, Mach numbers) *(Note: Dimensional Analysis is in PCB P-II U5 AND AEE Civil P-III U6, but NOT in AEE Common P-II U2!)*

SUBJECT 15B • HYDRAULIC MACHINES, CONTROL VALVES & HYDRO POWER PLANTS

AEE COMMON P-II UNIT 2 (PART 2) — EXCLUSIVELY FOR AEE CIVIL & AEE PR (ABSENT IN PCB!)

2. Fluid Mechanics and Machinery (Part 2: Turbomachinery, Valves & Hydro-Electric Plants) AEE CIVIL + AEE PR P-II

Basics of turbo machinery & Hydraulic Turbines: Hydrodynamic force of jets on stationary and moving flat, inclined, and curved vanes, velocity diagrams, work done and efficiency. Hydraulic Turbines: Classification of turbines, Heads and efficiencies, impulse and reaction turbines, Pelton wheel, Francis turbine and Kaplan turbine – working proportions, work done, efficiencies, hydraulic design – Draft tube theory – functions and efficiency. Performance of hydraulic turbines: Geometric similarity, Unit and specific quantities, characteristic curves, governing of turbines, selection of type of turbine, Cavitation.

Centrifugal Pumps: Classification, working, work done – barometric head – loss and efficiencies – specific speed – performance characteristic curves, NPSH. Selection of pumps and economic evaluation of pumping.

Hydraulic Directional & Pressure Control Valves: Hydraulic Directional Control – Check Valves, Shuttle Valves, two-, three- and four-Way Directional Control Valves, Directional Control Valve Actuation. Hydraulic Pressure Control – Pressure Relief Valves, Unloading Valves, Pressure Reducing Valves, Sequence Valves, Counterbalance Valves, Pressure Compensated Pumps.

Hydro Projects & Hydro Electric Power Plant: Classification – Typical layouts – plant auxiliaries – plant operation, pumped storage plants. Hydro Electric Power Plant: Water power – storage and Pondage – Hydrological cycle / flow measurement – drainage area characteristics – Hydrographs – classification of dams and spillways.

Module in AEE Common P-II U2	Core Sub-Topics for AEE Civil & AEE PR/TW Paper-II	Exam Separation & Hidden Trap Alert
1. Impact of Jets, Turbines & Centrifugal Pumps	• Jet impact on stationary/moving vanes & Velocity triangles • Pelton (Impulse), Francis & Kaplan (Reaction) Turbines, Draft Tube, Unit quantities (Nu, Qu, Pu), Specific speed (Ns), Cavitation & Thoma's σ • Centrifugal Pumps: Manometric head, Efficiencies, Minimum starting speed, Specific speed, NPSH	• **100% Absent in AEE PCB!** PCB Unit 5 stops at Boundary Layer and excludes all Turbomachinery. • High-weightage numerical zone (15–20 questions) in AEE Civil & PR Paper-II.
2. Industrial Fluid Power & Control Valves	• Directional Control Valves (DCVs): Check valve, Shuttle valve, 2/2, 3/2, 4/2, 4/3 way valves & Actuation methods (Manual, Pilot, Solenoid) • Pressure Control Valves (PCVs): Pressure Relief, Unloading, Pressure Reducing, Sequence & Counterbalance valves; Pressure-compensated pumps	• ⚠️ **HIDDEN MECHANICAL TRAP IN PAPER-II:** Most Civil textbooks omit Hydraulic Control Valves (DCVs/PCVs)! Prepare a 2-page notes sheet on ISO symbols & functions of these 7 valves to score 3–4 bonus marks in Paper-II!
3. Hydro Electric Power Plants, Hydrology & Dams	• Base-load vs Peak-load, Run-of-river vs Storage vs Pumped Storage plants, Storage vs Pondage, Surge Tanks, Penstocks, Forebay • Hydrological cycle, Stream flow measurement, Drainage basin characteristics, Hydrographs, Classification of Dams & Spillways	• **Direct Double-Benefit Overlap with AEE Civil Paper-III Unit 6!** Reading Hydrology, Hydrographs, Dams, and Spillways for Civil Paper-III automatically covers the final paragraph of Common Paper-II Unit 2 (giving AEE PR aspirants a Civil overlap edge too).

SUBJECT 23 • HYDRAULICS, HYDROLOGY, IRRIGATION & WATER RESOURCES

AEE CIVIL P-III UNIT 6 | OVERLAPS WITH COMMON P-II U2 & PCB P-II U5

6. HYDRAULICS AND WATER RESOURCE ENGINEERING (AEE Civil Paper-III Unit 6)

AEE CIVIL PAPER-III (U6)

Open Channel Flow (OCF): Open Channel flow – types of flows – Type of channels – Velocity distribution – Energy and momentum correction factors, uniform flow and calculation of uniform flow (Chezy's & Manning's), most economical section, Specific energy, critical flow conditions, critical depth computation, Non-Uniform flow: Assumptions and Equation for Gradually Varied Flow (GVF), types of channel bottom slopes, classification of surface profiles (M, S, C, H, A). Rapidly Varied Flow (RVF), hydraulic jump, energy dissipation.

Hydraulic Similitude: Dimensional analysis – Rayleigh's method and Buckingham's pi theorem – Geometric, kinematic and dynamic similarities – dimensionless numbers – model and prototype relations. Distorted and non-distorted models.

Hydrology & Floods: Hydrological cycle and its components, Precipitation and related data analysis, Evaporation and transpiration; S-hydrograph, Unit hydrographs. Floods and their management, Probable Maximum Flood (PMF); Streams and their gauging; Routing of floods; Capacity of reservoirs.

Irrigation & Hydraulic Structures: Multipurpose uses of water; Soil-Plant-Water relationships, Irrigation systems, Water demand assessment (Duty, Delta, Base period); Storages and their yields, Ground water yield and well Hydraulics (Darcy, Thiem, Dupuit); Water logging, drainage design. Design of rigid boundary canals, Lacey's and Kennedy's / tractive force concepts in canal design, Lining of Canals, Sediment transport in canals, Non-overflow and overflow section of gravity dams and their design, Energy dissipaters, tail water rating; Design of head works (Khosla's/Bligh's theory), Distribution works, Falls, Cross-drainage works, Outlets, River training.

Overlapping Topics (Read Once Across Papers!)	Exclusive Pure Civil Topics (ONLY in AEE Civil Paper-III Unit 6)	Exclusive Irrigation & Structures (ONLY in AEE Civil Paper-III Unit 6)
• **Dimensional Analysis & Similitude** (Rayleigh, Buckingham's Pi, Froude/Reynolds model laws): Common to **AEE Civil P-III U6 + AEE PCB P-II U5**! • **Hydrological Cycle, Stream Gauging, Drainage Basin, Hydrographs, Dams & Spillways:** Common to **AEE Civil P-III U6 + AEE Common P-II U2 (Civil & PR)**! • **Energy & Momentum Correction Factors (α, β):** Tested in both Pipe Flow (P-II) and OCF (P-III).	• Open Channel Flow (OCF): Chezy/Manning Uniform flow, Most Economical Sections (Rectangular, Trapezoidal, Circular), Specific Energy curve, Critical Flow, GVF Dynamic Equation & 12 Surface Profiles, RVF & Hydraulic Jump • Distorted vs Undistorted Hydraulic Models • Advanced Hydrology: Raingauge network, Missing rainfall, Infiltration (φ & W indices), Unit Hydrograph derivation, S-Curve, Muskingum Flood Routing, Mass Curve & Reservoir Capacity	• Irrigation Water Requirement: Field Capacity, Permanent Wilting Point, Consumptive Use, Duty-Delta-Base Period, Irrigation Efficiencies • Well Hydraulics (Confined/Unconfined aquifers) & Tile Drainage / Waterlogging • Alluvial Canal Design (Kennedy's & Lacey's Regime theories, Tractive force) • Gravity Dams (Elementary profile, Uplift, Stability), Diversion Headworks (Bligh, Lane, Khosla), Canal Falls, Cross-Drainage (Aqueduct, Syphon, Superpassage) & Outlets

SUBJECTS 16 & 17 • ENGINEERING MATERIALS & PROCESS CALCULATIONS / THERMODYNAMICS

AEE PCB PAPER-II UNIT 4 & UNIT 6 (100% EXCLUSIVE TO AEE PCB — ABSENT IN CIVIL & PR!)

4. ENGINEERING MATERIALS (AEE Notification 08/2026 — AEE PCB Paper-II Unit 4) AEE PCB PAPER-II (U4)

Structure and properties of engineering materials, phase diagrams, heat treatment, stress-strain diagrams for engineering materials. *(Note: This is Materials Science / Metallurgy, NOT Civil Building Materials like Bricks/Timber/Cement!)*

6. PROCESS CALCULATIONS AND THERMODYNAMICS (AEE PCB Paper-II Unit 6) AEE PCB PAPER-II (U6)

Laws of conservation of mass and energy, use of tie components; recycle, bypass and purge calculations; degrees of freedom, First & Second law of thermodynamics and their applications; equations of state and thermodynamic properties of real systems.

PCB Exclusive Unit	Exact Micro-Topics to Study ONLY for AEE PCB Paper-II	Boundary Rule (What NOT to Read!)
Subject 16: Engineering Materials (PCB Unit 4)	• Crystal Structures (BCC, FCC, HCP: Packing factor, Coordination number, Miller indices, Crystal defects: Point, Line/Dislocations, Surface) • Mechanical Properties & Stress-Strain Diagrams (Elastic/Plastic deformation, Strain hardening, Creep, Fatigue, Hardness tests: Brinell, Rockwell, Vickers) • Phase Diagrams: Gibbs Phase Rule (P + F = C + 2), Lever Rule, Isomorphous & Eutectic systems, Iron-Carbon (Fe-Fe3C) Equilibrium Diagram (Austenite, Ferrite, Cementite, Pearlite, Ledeburite, Martensite; Eutectoid, Eutectic, Peritectic reactions) • Heat Treatment of Steels: TTT Diagram, Annealing, Normalizing, Hardening, Tempering, Case Hardening (Carburizing, Nitriding, Cyaniding)	• **Do NOT confuse with Civil Unit 1 (Building Materials):** PCB Unit 4 is Mechanical/Metallurgical Material Science. • **Only Overlap with Civil/PR:** Stress-strain diagram & mechanical properties (Strength, Ductility, Toughness, Resilience, Malleability).
Subject 17: Process Calculations & Thermodynamics (PCB Unit 6)	• Chemical Process Stoichiometry: Mole/Mass fractions, Limiting & Excess reactants, Conversion, Yield, Selectivity • Material & Energy Balances (Steady-state with and without chemical reaction) & Use of Tie Components (Inert balance) • Recycle, Bypass, and Purge Stream Calculations (Purge ratio to prevent inert buildup) & Degrees of Freedom Analysis (N_DF = N_variables − N_equations) • 1st Law (Closed & Open systems/SFEE, Internal Energy, Enthalpy) & 2nd Law of Thermodynamics (Carnot cycle, Entropy, Clausius inequality, Gibbs & Helmholtz free energy) • Equations of State (Ideal Gas, Van der Waals, Redlich-Kwong, Virial, Compressibility factor Z) & Maxwell Relations / Real gas properties	• **Do NOT import Mechanical Paper-III (IC Engines, Boilers, Refrigeration, Gas/Steam Turbines):** Stick strictly to Chemical/Process Stoichiometry (Recycle/Bypass/Purge/Tie component) + Basic 1st/2nd Law & Equations of State!

🎯 **MASTER STUDY SEQUENCE FOR CORE ENGINEERING PAPER-II (CIVIL vs PR vs PCB):**
Step 1 (Triple-Overlap Fluid Mechanics First): Study *Fluid Properties, Manometers, Kinematics, Bernoulli, Momentum, Pipe Flow, Boundary Layer & Dimensional Analysis*. This single pass serves **AEE Civil (P-II & P-III), AEE PR/TW (P-II), and AEE PCB (P-II Unit 5)**.
Step 2 (AEE Civil + AEE PR Paper-II Block): Complete *SOM (Unit 1) + Hydrostatics/Buoyancy + Flowmeters + Turbomachinery + Hydraulic Control Valves + Hydro Power Plants (Unit 2)*. Since Civil and PR share the exact same Paper-II, this locks in **150 Marks for both exams simultaneously**.
Step 3 (AEE PCB Exclusive Block): Dedicate 4–5 focused days to **Crystal Structures, Fe-C Phase Diagram, Heat**